

DocStock

Reminder a day keeps Doc at bay

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Team Pearl



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Isaiah Gamble is a senior at ODU majoring in Computer Science with a minor in Mathematics. He is a Coast Guard veteran and has "Experience Every Natural Disaster" on his bucket list. He plans on pursuing the arts after completion of his degree.

Julia Hairston is a senior at ODU majoring in computer science. She hopes to pursue a master's in data science in the future. She enjoys working out, listening to music, and doing puzzles in her free time.



Julia Hairston

Team Pearl



Shannon Stinnette

Shannon Stinnette is a senior at ODU majoring in Computer Science. While working part-time as a server, she is currently working towards a variety of networking certifications. Other than working she spends her free time enjoying video games, books, and hanging out with friends.

Anson Cheng is a senior at ODU majoring in Computer Science, hoping to pursue a Master's. He enjoys playing guitar, gaming with friends, and wracking his head on problems.



Anson Cheng

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Ab Aljazaeri

Abdelrahman "Ab" Aljazaeri is a senior at Old Dominion University majoring in Computer Science with a minor in Mechanical Engineering Technology. He likes to tinker with cars and dabbles in network security. He plans to keep working where software meets hardware after graduation.

Toni Sallaku is a senior athlete at ODU studying Computer Science. He is a swimmer, Lifeguard and Water Safety Instructor at the SRC. He likes working out, playing sports, and playing videogames.



Toni Sallaku

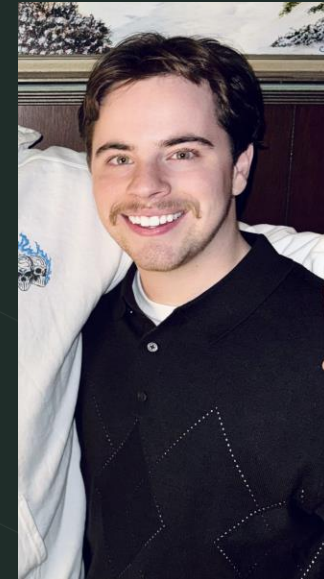
Team Pearl



Jessica Fischer

Jessica Fischer is a junior at Old Dominion University in the Linked Masters program for Computer Science. She enjoys gaming, working out, and hanging out with people in her free time.

Matthew Richards is a Senior Computer Science major at ODU. After graduating in May, he plans to move to Arlington, VA, and work for CACI doing backend development. He enjoys playing soccer, gaming, and playing the piano in his free time.



Matthew Richards

Problem Statement

Many people who take medications regularly have difficulty managing their medication schedules, remembering doses, tracking whether medications were taken, and knowing if their supply is running low. These difficulties can lead to missed, delayed, or repeating doses of medicine already taken.

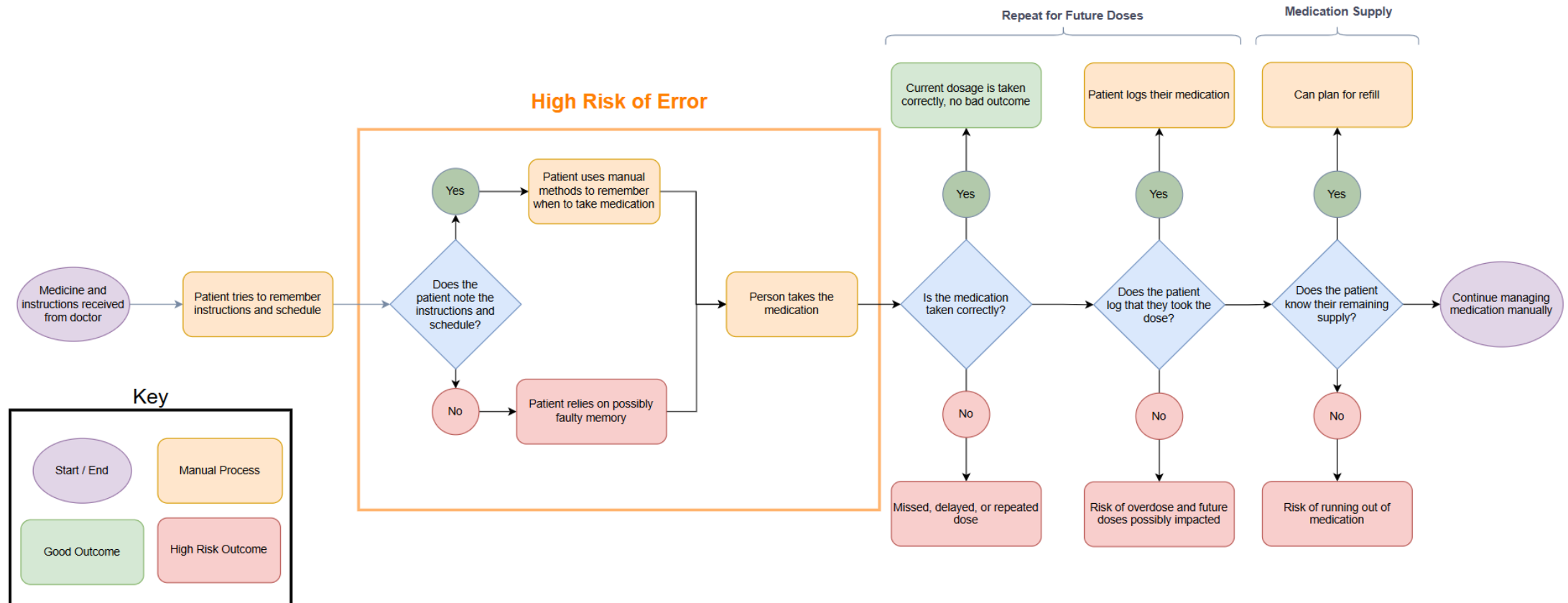
Problem Characteristics

According to the Centers for Disease Control and Prevention (CDC), about one in five new prescriptions in the United States are never filled.

Approximately 50% of filled prescriptions are taken incorrectly with regards to timing, dosage, frequency, and duration.

Direct health care costs associated with nonadherence have grown to approximately \$100-\$300 billion of U.S. health care dollars spent annually. [1]

Current Process Flow





Elevator Pitch

DocStock is a web and mobile app that helps people and their care givers manage prescriptions by scheduling doses, screening for drug interactions, and tracking supply, with the goal of increased medication adherence.

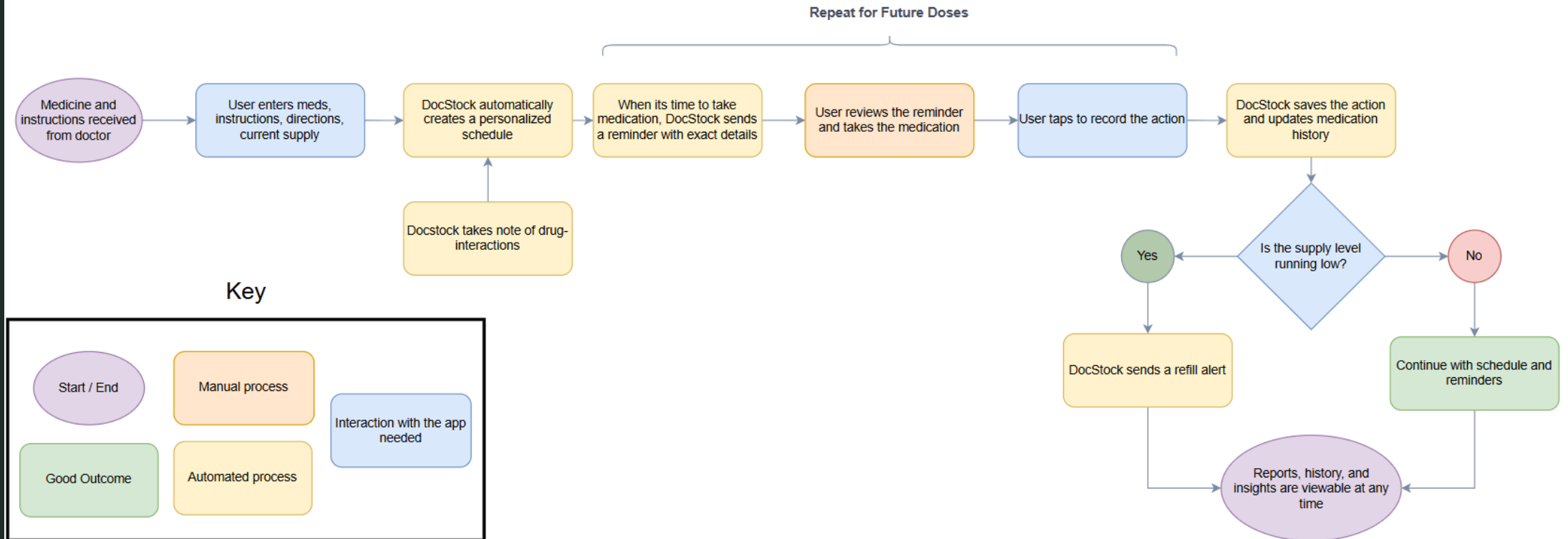
Solution Statement

DocStock will be a smartphone and web application that helps people manage their prescribed medication schedules. Users enter their medications, dosing instructions, and current supply, and DocStock organizes everything into a single, easy-to-follow schedule.

When a new medication is added, DocStock checks it against the user's active medication list and warns the user of known drug interactions. The app sends reminders when doses are due, lets users record each dose taken, tracks remaining supply, and alerts the user before a refill is needed.

Adherence history is compiled into a report that can be exported for a healthcare provider. Caregivers can be granted access, and multiple profiles can be managed under one account.

Solution Process Flow



The number of manual processes is reduced to one compared to the current process flow.



What it Will Do

- Centralize every prescription in one dashboard reachable from any device.
- Schedule doses and send reminders so doses aren't missed, mistimed, or doubled
- Alert users before refills run out to prevent gaps in treatment
- Warn of drug interactions by checking each newly added medication against the user's active list
- Track adherence and export a provider-ready report to bring to appointments
- Share access with caregivers and support multiple profiles under one account

What it Will Not Do

- It will not automatically order refills for low medications.
- It will not give medical advice or replace healthcare professionals. Interaction warnings are informational only; users are directed to confirm with their provider or pharmacist.
- It will not log symptoms, mood, or health metrics (blood pressure, glucose, etc.)
- It will not integrate directly with pharmacies to pull prescription automatically.

Competition Matrix

Feature	DocStock	Medisafe [2], [3]	MyTherapy [4]	Round Health [5]
Dose reminder & scheduling	✓	✓	✓	✓
Prescription refill alert	✓	✓	✓	✓
Drug interaction warning	✓	✓	-	-
Caregiver / family sharing	✓	✓	✓	✓
Exportable adherence report	✓	Partial	✓	-
Symptom / health metric logging	-	-	✓	-
Multiple user profiles	✓	✓	✓	✓
Web-based (browser and phone app)	✓	-	-	-
Free / no paywall	✓	Partial	✓	Partial

Development Tools

- IDE: Visual Studio Code. Writing and editing code.
- Web Frontend: HTML, CSS, JavaScript. Website interface.
- Mobile Frontend: React Native. Mobile application.
- Backend: Node.js & Express. Handle server-side logic and API requests.
- Database: MySQL. Store users, medications, schedules, history, etc.
- Testing: React Native Testing Library. Test frontend/backend functionality.
- Continuous Integration & Continuous Deployment: GitHub & GitHub Actions. Automated testing and version control

Major Functional Components

Web Application: HTML, CSS, JavaScript

- HTML for the website's structure

- CSS for styling

- JavaScript for interactivity

Mobile Application: React Native

- React Native will be used for both iOS and Android development

Backend: Node.js, Express.js

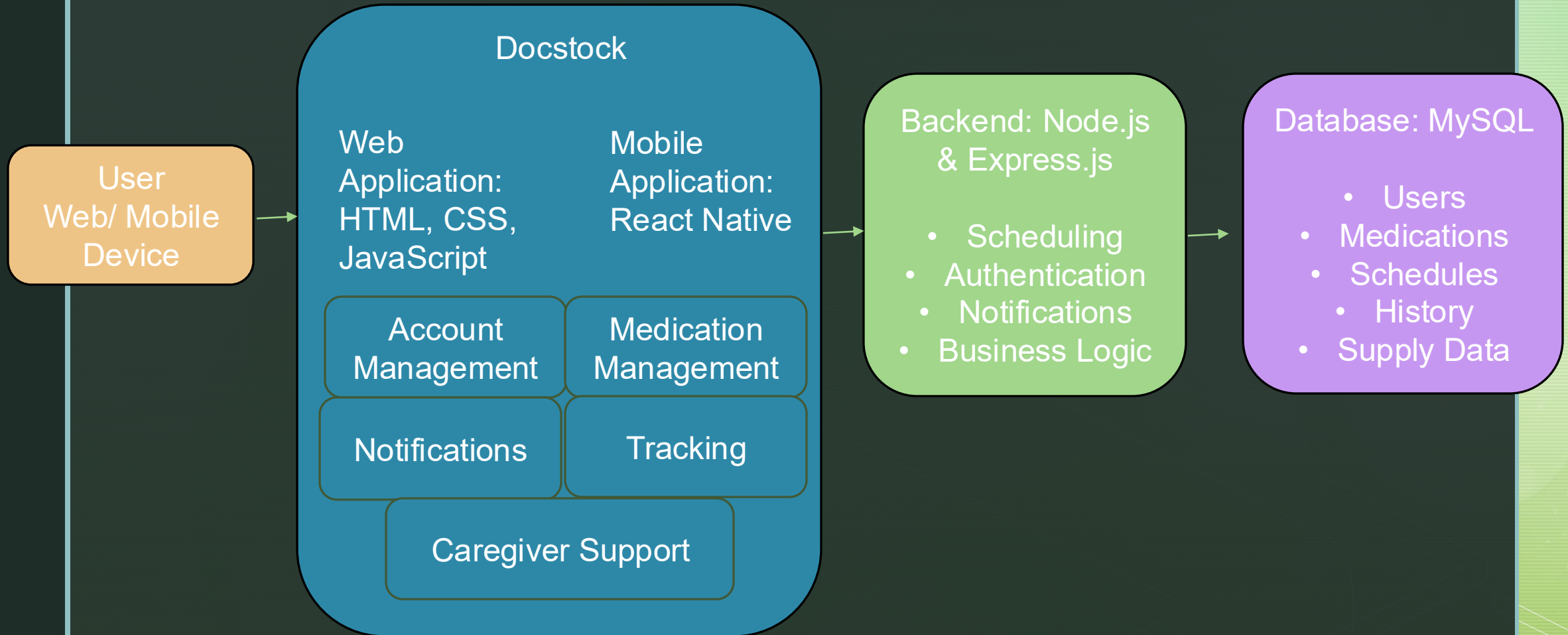
- Node.js will be the server-side environment

- Express.js will be for backend and APIs

Database: MySQL

- MySQL will store and manage user information

Major Functional Components Diagram



Customer and End User Risks

Risk	Probability	Impact	Mitigation
User forgets to enter or update their medications	3	5	The app will use a simple set-up process with clear prompted instructions for adding and updating a medication. There will be a clearly visible button to add medication.
User ignores or misses medication reminders	2	5	Provide regular notifications and an easy way to record having taken a medication.
The app is difficult for some users to navigate	4	4	Large text, obvious buttons, simple navigation. User testing can identify problems before actual release.
User enters incorrect medication info	3	5	The app will require confirmation of medication details. It will clearly state that users should follow instructions provided to them by their healthcare providers.

Technical Risks

Risk	Probability	Impact	Mitigation
Delayed or failed notifications	2	5	Test notifications across supported devices. Provide in-app schedule for users to manually track. Alert users they must turn on app notifications to get them.
Smartphone compatibility problems	3	5	Test the application on multiple supported devices and OS versions via emulators.

Security Risks

Risk	Probability	Impact	Mitigation
An unauthorized user gains access to medication information	2	3	App will be password protected
An unauthorized user gains access to user's username and password	2	4	App will provide users with an optional two-step security verification to add an additional layer of account security [6]
User forgets username or password	4	2	App will allow users the ability to change username and password

Legal Risks

Risk	Probability	Impact	Mitigation
User claims the app caused harm because they relied on a reminder.	3	5	Communicate the app's limitations and advise users to verify medication instructions with their healthcare provider.
App unintentionally infringes on copyright or trademarks	2	3	App will use original or licensed materials with proper credit.

References

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Appendix

